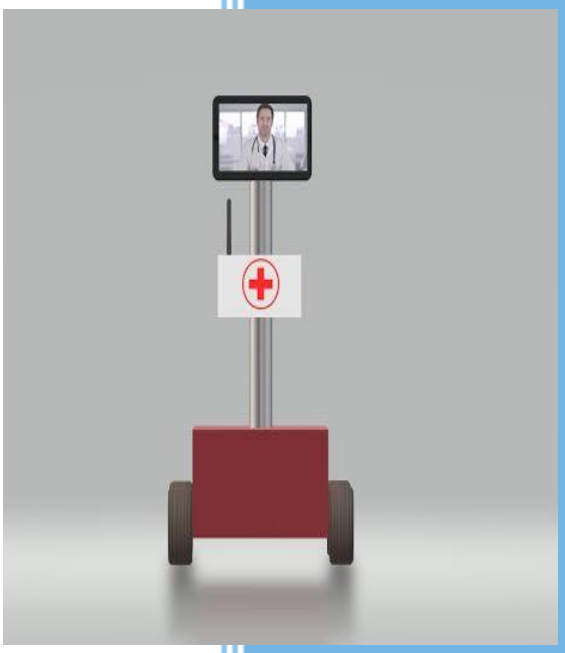
2023/2024

Medical Assistant Robot



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Misr international technical university

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(Chapter one)

1.1 Introduction

The medical assistant project Is so important in our live because is used to help patients to know what your diagnosis is have and what are initial treatment for you. when the patient enters your symptoms, the server will out for you the initial diagnosis and initial treatment for his disease. We will design circuit for hardware part of project. The car circuit is had some of components like: Arduino uno to control to complete process for this circuit, motor drive shield to install components for this circuit on that, TT gear motor (*4), wheels (*4), servo motor, ultrasonic sensor to control on motivation, increased sensor (*2), 18650 Li-on battery (*2), DC power switch, battery holder. We make this car circuit to go to patient when need that to can enter your symptoms is have to obtain an initial diagnosis based on his symptoms.

This project is very simple and useful for patients and doctor, helps patients when enter the symptoms your have and get for you the initial diagnosis based on his symptoms and write for them initial treatment for his disease until they are admitted to the doctor.

The medical assistant project is had two parts (software (AI), and hardware). In software part, we will collect data set is have some of symptoms and diagnosis and some initial treatment for some of diseases, then we will make cleaning for this dataset to prepare that to make processes on that. Next, we will create 3 types of algorithms like: decision tree, naïve bays, and random forest. These algorithms used to select what is disease for the patient and select the suitable initial treatment for you.

(Chapter two)

2.1problem definition and characteristics of medical assistant project:

1-environmental issues:

The project does not negatively affect the environment, is very friendly to the environment, and helps me in practical life because it's just a type of the security

2-relability:

I can make this project because it's easy to instruct and the components for project is available on market and the software it helps patients to know what your diagnosis and the initial treatment for them.

3-operation condition:

Operate by 220 v adapter

4-cost:

3000L.E

5-Size:

15.0cm*25.8cm*39.8cm

6-physical resources:

- TT Gear motor
- Arduino Uno
- Wheels (*4)
- Battery holder
- Li-on battery (*2)
- Ultrasonic sensor (HC-SR04)
- DC power switch
- Motor drive shield
- Wire male/female
- Wire male/ male

7-software resources:

- Visual studio code
- Arduino IDE
- Tinker cad simulation software
- Excel, word, power point micro soft

(Chapter three)

3.1The project procedures

A- Description maker (team leader) (Mosab Mohamed Abd El Ghani) he is the leader of team, he is distributing roles. And he is specifying the time of meeting.

B- Meeting holder (sub leader) (Rana Mohamed Farouk) she is helping the leader, she is replacing the leader if he isn't here, also helping leader in meetings

C- budget holder (Mohammed Omar Sultan, Nora Mahmoud) They are responsible for collecting money of project from the team work, they responsible to go to buy components of project.

D-collect and cleaning data (Mosab Mohamed Abd El Ghani, Rana Mohamed Farouk, Mohammed Omar Sultan) we make cleaning for data to prepare that to make some preprocessing on this data set.

E- create three algorithms (decision tree algorithm, naïve bays algorithm, and random forest algorithm) (Mosab Mohamed Abd El Ghani, Mohammed Omar Sultan, Omar Amin Gaber) create these algorithms to used that in know or select the initial diagnosis for patients and select for patients the initial treatment for them.

F-web site part (Mosab Mohamed Abd El Ghani, Nora Mahmoud Abd El hakim, Mohamed Omar Sultan, Muhammed Sha`ban Muhammed, Omar Amin Gaber, Rana Mohamed Farouk) we create web site to show on that for patients some symptoms, diagnosis, and some initial diseases and initial treatment for patients

G- connect the project (Mosab Mohamed Abd El Ghani, Rana Mohamed Farouk, Nora Mahmoud Abd El hakim, Mohammed Omar Mohammed, Mohamed Gamal Mohamed, Ghada Ahmed Al Amir, Omar Amin Gaber, Ahmed Amir Salem, Mohamed Sha`ban) We meet to know how to make the project, how to connect components and products of the project and study the positive and negative aspects of the final project.

H- Reporting the project (Mosab Mohamed Abd El Ghani, Rana Mohamed Farouk, Nora Mahmoud Abd El hakim, Mohammed Omar Sultan, Mohammed Gamal Mohammed, Ghada Ahmed Al Amir, Omar Amin Gaber, Ahmed Amir Salem, Muhammed Sha`ban Muhammed) We all are responsible for writing the reports of the project and each one of us responsible of part of the project.

I- making presentation (Mosab Mohamed Abd El Ghani, Rana Mohamed Farouk, Nora Mahmoud Abd El hakim, Mohammed Omar Sultan, Mohammed Gamal Mohammed, Ghada Ahmed Al Amir, Omar Amin Gaber, Ahmed Amir Salem, Muhammed Sha`ban Muhammed) we meet to discuss how to present the presentation, then we start presenting it and distribute the roles to the team members

J- Make a plan (Mosab Mohamed Abd El Ghani, Rana Mohamed Farouk, Nora Mahmoud Abd El hakim, Mohammed Omar Sultan, Mohammed Gamal Mohammed, Ghada Ahmed Al Amir, Omar Amin Gaber, Ahmed Amir Salem, Muhammed Sha`ban Muhammed.) We meet to organize the project ideas and management and plan for the project completion date.

K-Writing report (Mosab Mohamed Abd El Ghani, Rana Mohamed Farouk, Nora Mahmoud Abd El hakim, Mohammed Omar Sultan, Mohammed Gamal Mohammed, Ghada Ahmed Al Amir, Omar Amin Gaber, Ahmed Amir Salem, Muhammed Sha`ban Muhammed). We meet for writing the project how it operates, its advantages and disadvantages, specification and problems that we found.

3.2 The circuit diagram of car circuit

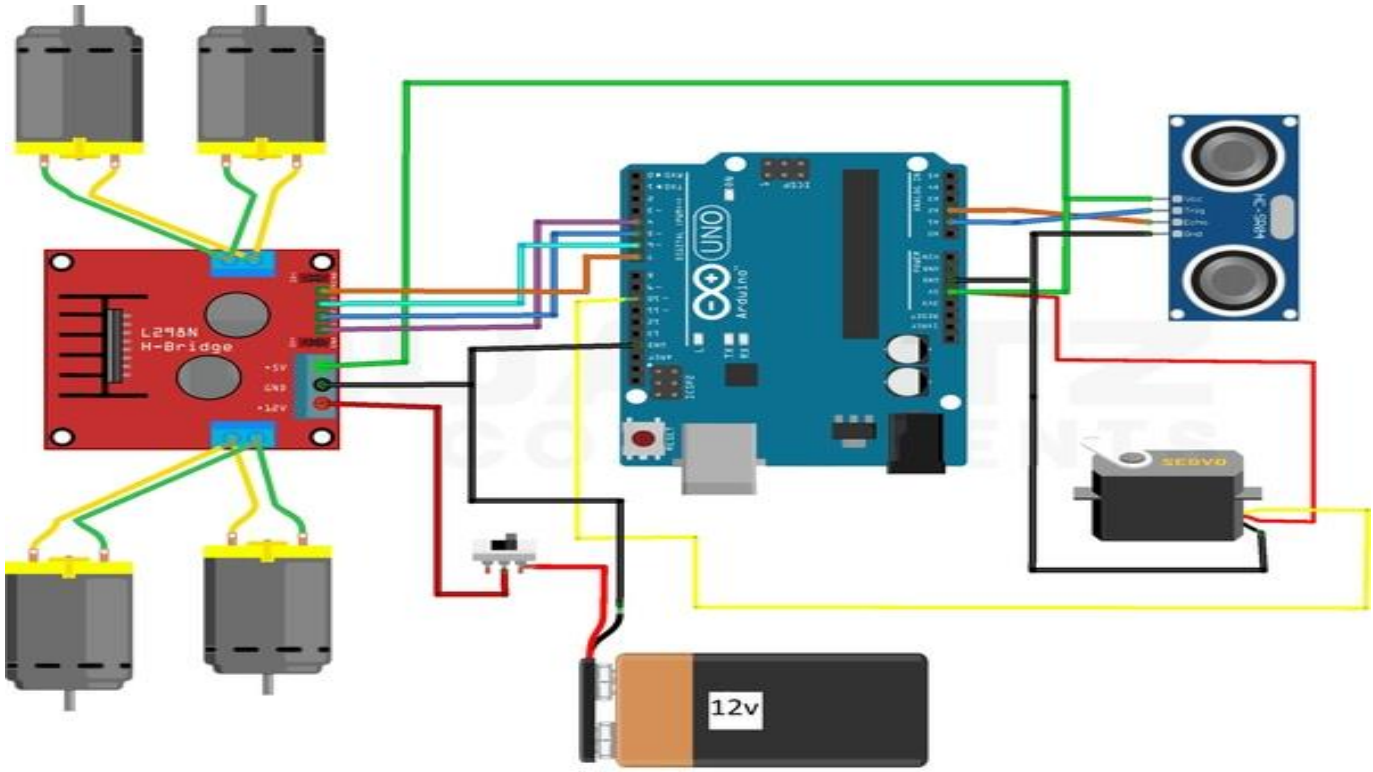


Figure 1 circuit diagram

3.3 Software package

-visual studio code

-tinker cad simulation program

-Arduino IDE

-Excel

-Power point

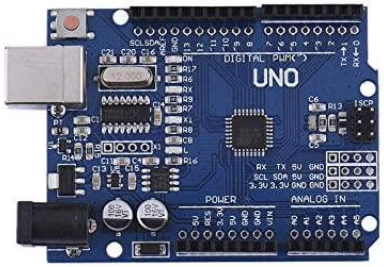
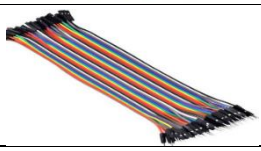
-Word

-Project text book

-Photo viewer

-Video player

3.4 Requirement recourse

Resource		Picture
1	Motor drive shield	
2	Wheels (*4)	
3	Arduino Uno	
4	TT gear motor	
6	Ultrasonic sensor(HC-SR04)	
7	Wire male/female	

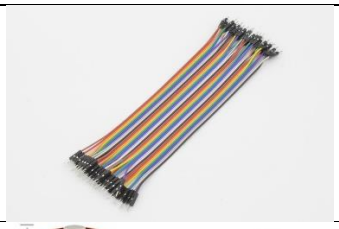
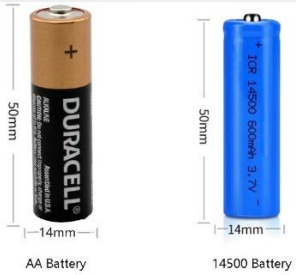
8	**Male-male wires**	
9	**18650 Li-on battery**	
10	**Holder battery**	
10	**DC power switch**	

Table 1 requirement components

(Chapter four)

Introduction of chapter4

In this chapter will talk about function for components for the medical assistant project circuit

1-arduino uno:

Arduino uno acts as a main control of electronic car circuit. That receives signals from different sensors and control the car based on those signals. Processing for Arduino uno reads these sensor signals and figures out what they mean. the output for Arduino uno is based on the sensor reading, the Arduino uno sends instructions to their parts of the car circuit. And it can control things in this car like: motor drivers (to regulate motor speed and direction). the Arduino uno takes in information from the car's environment through sensors and translates that information into actions that make the car move.

2-motor drive shield:

A motor driver shield in an electronic car circuit acts like an intermediary between the control (Arduino Uno) and the muscles (motors) of the car. Motor Driver Shield is a board which gives possibility to control 4 DC motors or 2 step motors (unipolar or bipolar) by on Arduino. And it is allowing for smooth control of your electronic car.

3-Ultrasonnic sensor (HC-SR04):

The ultrasonic sensor is a non-contact type of sensor used to measure an object's distance and velocity. This sensor operates on sound wave property to measure the velocity and distance of the object

4-Li-on battery:

the Li-ion battery is the electric car's energy source, allowing it to run and be recharged.

5-battery holder:

The primary function of a battery holder is to keep cells fixed in place safely and securely while conveying power from the batteries to the car circuit device.

6-TT Gear motor:

A TT motor is a type of DC motor that has a gearbox attached to it. The gearbox reduces the speed of the motor and increases its torque. used To control the speed and direction of the car

7-DC Power switch:

The DC power switch in an electronic car circuit acts like a master on/off switch for the entire electrical system.

8-wheels:

Wheels themselves aren't directly part of the electronic circuit in an electric car. The electric circuit powers the electric motor, which in turn spins a shaft connected to the wheels. So, the wheels are more like the mechanical output of the electrical circuit.

9-male-male wires:

male-male wires provide a quick and easy way to connect two male terminals on different components, but they're not a standard part of a car's permanent wiring.

10-female-male wires:

the function of these connectors is to simply and securely join two wires in the car's electrical system.

4.1 car circuit for medical assistant project

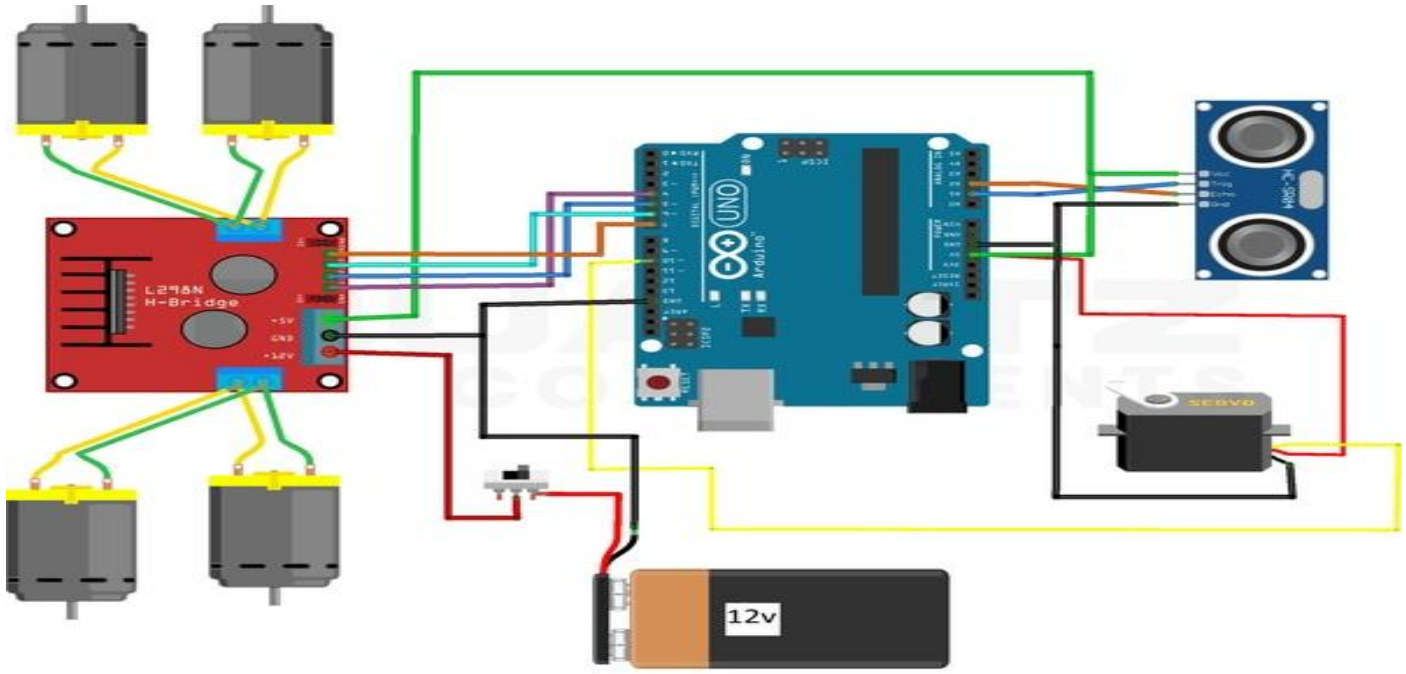


Figure 2 circuit diagram

4.2 Operation:

This circuit characterized by its ease because it has few components. The car circuit for medical assistant project is embedded system. So, this circuit isn't complicated. The operation of this system is the main idea for this circuit is converting stored electrical energy to motion. Arduino uno in this circuit is main controller is receiving the driver's input and adjusting the inverter to send more or less AC power to the motor, controlling the car's speed. The battery is the car's energy source. A motor driver shield in an electronic car circuit acts like an intermediary between the control (Arduino Uno) and the muscles (motors) of the car. will used TT Gear motor to control the speed and direction of the car. We used ultrasonic in this circuit to detect found patient or not when found patient will car go to patient. we used dc power switch in this circuit acts like a master on/off switch for the entire electrical system.

4.3. the characteristics for install circuits

1-cost:

3000l.e

2-life time:

10years

3-No. of components:

10 components

4-time of impairment project:

22 days

5-complexity:

Low

6-size:

15.0*25.8*39.8 centimeter

7-maintenance:

2 years

8-weight:

Medium

9-goals of project:

Reduce cost and danger

High performance and accurate

Increased availability and reliability

Steady, Unemotional and complete responses at all times

(Chapter five)

5.1 Gantt chart of project

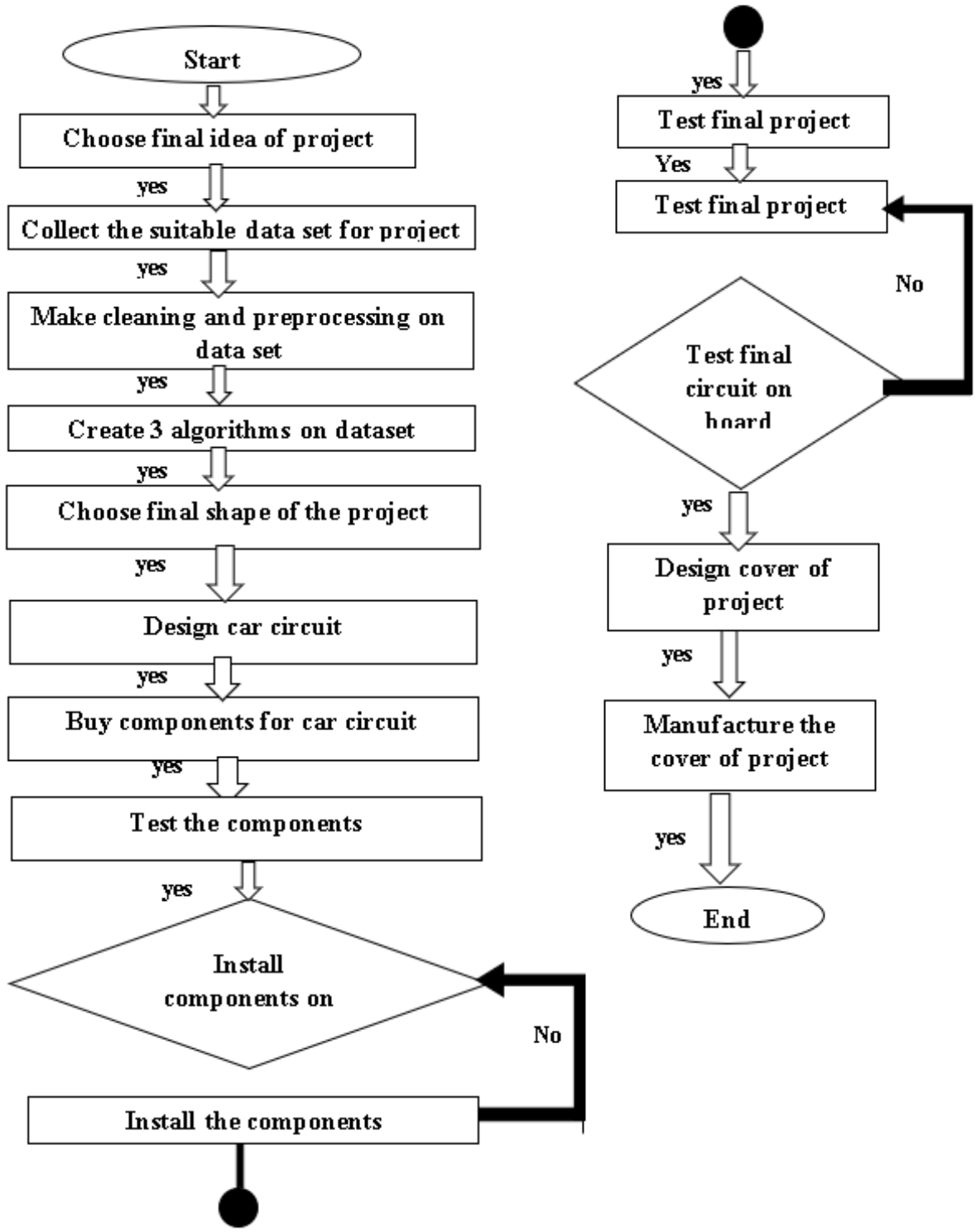
Table 2 Gantt chart

Task \ Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Choose the idea of the project	■																					
Collect suitable data set for project		■	■																			
Make cleaning and preprocessing on data set				■																		
Create 3 algorithms on data set					■	■																
Choose final shape of project							■	■														
Design car circuit									■													
Buy the components										■												
Test the components											■	■	■	■								
Install components on board															■							
Test the circuit																■	■					
Install the final circuit on board																		■	■	■	■	
Test the final circuit on board																						■
Design and manufacture the cover of project																						■

5.2 Explain of Gantt chart

1. **Choose the idea of the project:** in this point, we met together to discuss the project's ideas.
2. **Collect the suitable data set for project:** in this point, we collect data set is have some of symptoms, diagnosis, diseases, and treatments for theses diagnosis and diseases.
3. **Make cleaning and preprocessing on data set:** in this point, we make cleaning for data set and some preprocessing to prepare data to make some of algorism and processes on that.
4. **Create three algorithms:** in this point, we create 3 algorithms (random forest algorithm, decision tree, naïve bays) used that in select the initial diagnosis and initial treatments for patients.
5. **Choose final shape of project:** in this point we discuss the project's shape
6. **Design the circuit on tinker cad simulation program**: in this point, I connect the circuit of the project in the tinker cad.
7. **Buy the require components**: in this point, we went to the electronics shops (RAM) to buy the require component of the project.
8. **Test the component**: in this point, we test all the component before we connect it to ensure that all the component working.
9. **Test the circuit:** in this point, we test the connection of the circuit to make sure that the connection is correct.
10. **Install the final circuit on board:** in this point, we install the final circuit after testing the components and the connection to test the final circuit.
11. **Test the circuit on board:** in this point, we test the components and the final circuit connection to start putting it on the cover.
12. **Design the cover of the project:** in this point, we design the project cover shape and we choose the appropriate dimensions
13. **Manufacture the cover of the project:** in this point, we construct the cover of the project according to dimensions and weight.
14. **Put the final circuit on the cover:** in this point, we will put the final project in the cover and operate it.

5.3 Flow chart



5.4 Explain flow chart of the project

1- Choose the final idea of the project:

we will collect the ideas of my group (8 ideas) and explain them, then we vote for the best idea.

2-collect the suitable data for project:

We Collect data set is have some of symptoms, diagnosis, diseases, and treatments for this diagnosis.

3-make cleaning and preprocessing on data set:

We make cleaning and some preprocessing to prepare data set to make on that any processing and create algorithms on that

4-create three algorithms on data set:

We create three algorithms on data set(naïve bays algorithm, decision tree algorithm, random forest algorithm) to select and help patients to know what the initial diagnosis is have and the initial treatment for this diagnosis.

5-choose the final shape of project:

We discuss the final shape for this project

6- Design the car circuit on tinker cad:

I will design the circuit on tinker cad software to ensure that the circuit work.

7- Buy the component of the project:

We will buy the component as a group.

8- Test the component of project:

we will test the components to make sure they work or not.

- If all the components are working, I will plug them on the PCB.

- If no component works, I will buy another working component.

9- install the component on test board:

After testing the components and making sure all the components are working fine then I connect them on the test board.

10-Test the circuit on the test board:

After connect the component on the test board as a circuit we will test it

11- install the final circuit on the bread board:

In this point we will connect the final circuit on the bread bord

12-Test the circuit on the bread board:

I will test the project to make sure that the project work.

-If the project works well, I will install the final shape.

-If it doesn't work, I'll go back to the previous step and make sure every component is well connected.

13- design the cover:

After testing the project and making sure it works, I'm going to design the cover of the project and choose the appropriate dimensions to manufacture it.

14- Manufacture the cover of the project:

We will send the design of the cover to a specialized place to manufacture it in the required sizes and color it black.

5.5 Physical and human resources

We used the tinker cad circuit simulation program to test the circuit before starting to connect it in the lab, that we start to test the circuit in the IT laboratory and the measurement lab with the help of the assessor DR/ Marwa we used the test board and the multimeter to measure the volt, current, and resistance on the components of the circuit. Then we start to connect the circuit by using the caustic and tin. We used Microsoft Office PowerPoint to make the presentation of the project.

(Chapter six)

6.1 Design the circuit on the simulation program

To design the car circuit of medical assistant project, we first draw the car circuit on a simulation program (tinker cad) to ensure that the circuit is working well.

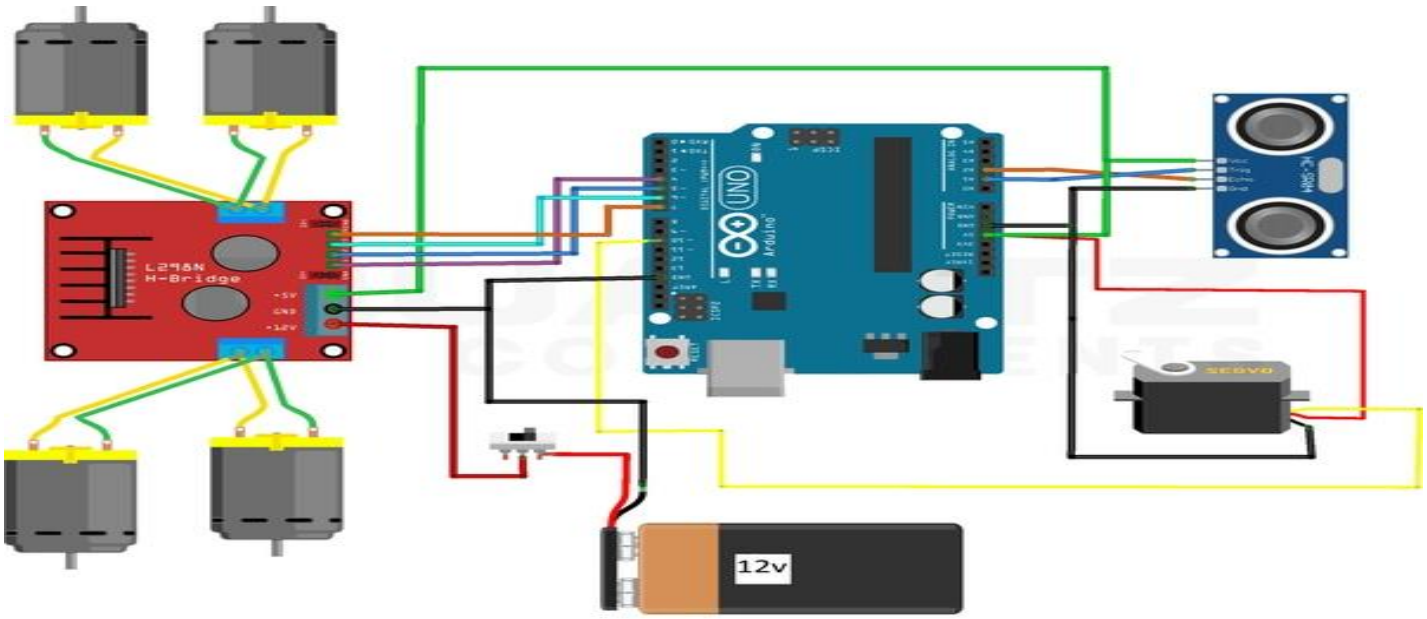


Figure3. simulation circuit

6.2 Implantation:

The car circuit for medical assistant project is embedded system. The operation of this system is the main idea for this circuit is converting stored electrical energy to motion. Arduino uno in this circuit is main controller is receiving the driver's input and adjusting the inverter to send more or less AC power to the motor, controlling the car's speed. The battery is the car's energy source. A motor driver shield in an electronic car circuit acts like an intermediary between the control (Arduino Uno) and the muscles (motors) of the car. will used TT Gear motor to control the speed and direction of the car. We used ultrasonic in this circuit to detect found patient or not when found patient will car go to patient. we used dc power switch in this circuit acts like a master on/off switch for the entire electrical system.

6.3 The Components needs to implant this project / Material Required:

1-Arduino Uno: Arduino uno acts as a main control of electronic car circuit. That receives signals from different sensors and control the car based on those signals. Processing for Arduino uno reads these sensor signals and figures out what they mean. the output for Arduino uno is based on the sensor reading, the Arduino uno sends instructions to their parts of the car circuit. And it can control things in this car like: motor drivers (to regulate motor speed and direction). the Arduino uno takes in information from the car's environment through sensors and translates that information into actions that make the car move.

2-motor drive shield: A motor driver shield in an electronic car circuit acts like an intermediary between the control (Arduino Uno) and the muscles (motors) of the car. Motor Driver Shield is a board which gives possibility to control 4 DC motors or 2 step motors (unipolar or bipolar) by on Arduino. And it is allowing for smooth control of your electronic car.

3-Ultrasonnic sensor (HC-SR04): The ultrasonic sensor is a non-contact type of sensor used to measure an object's distance and velocity. This sensor operates on sound wave property to measure the velocity and distance of the object.

4-Li-on battery: the Li-ion battery is the electric car's energy source, allowing it to run and be recharged.

5-battery holder: The primary function of a battery holder is to keep cells fixed in place safely and securely while conveying power from the batteries to the car circuit device.

6-TT Gear motor: A TT motor is a type of DC motor that has a gearbox attached to it. The gearbox reduces the speed of the motor and increases its torque. used to control the speed and direction of the car.

7-DC Power switch: The DC power switch in an electronic car circuit acts like a master on/off switch for the entire electrical system.

8-wheels: Wheels themselves aren't directly part of the electronic circuit in an electric car. The electric circuit powers the electric motor, which in turn spins a shaft connected to the wheels. So, the wheels are more like the mechanical output of the electrical circuit.

6.4 Implant steps:

1) we wrote the programming code of the system on the Arduino IDE and compiled and check it form any errors before burning it on the Arduino.

2) We start to connect the according to the circuit diagram of the system.

3) We Use lines to represent wires connecting the components. Label the lines with voltage values if necessary (5V, GND).

4) The battery positive (+) typically we connect to the Arduino's Vin pin. The negative (-) connects to the Arduino's GND pin.

5) The DC power switch we can connect between the battery positive (+) and the circuit to control power.

6) The ultrasonic sensor when will have specific connections based on the model. Connect the sensor's "VCC" pin to the Arduino's 5V pin, Connect the sensor's "GND" pin to the Arduino's GND pin, Connect the sensor's "Trig" pin to an Arduino digital output pin (pin 8), and connect the sensor's "Echo" pin to an Arduino digital input pin (pin 7).

7) The motor driver shield we connects to the Arduino's digital pins for motor control signals and a PWM pin for speed control.

8) The 4TT gear motor we connect to the motor driver shield based on its specific outputs (e.g., IN1, IN2).

9) We connect the USB cable of the Arduino to the laptop and the USB port on the Arduino to burn the code.

10) We finally connect the adapter cable to the power port on the Arduino.

11) We put the final circuit on the cover of the project after we ensure that the project is working.

12) Finally marketing our project

6.5 Similarly of the project:

The medical assistant project is consisting for two parts (software part (AI) and hardware part). The software part is consisting of data set that contains some of the diseases, symptoms, and initial treatment for these diseases. And through this data set we create three algorithms (random forest, decision tree, naïve bays) theses algorithms are used to help doctor and patients to identify diseases and diagnosis and the initial treatment for these patient's symptoms. The hardware part is consisting of electric car circuit model is have some of components like: the main control components for this circuit is Arduino UNO, motor drive shield, 4 TT Gear motor, ultrasonic sensor (HC-SR04),4 wheels, Li-on battery, battery holder, DC power switch.

6.6 Connection and project photos:

When programming the Arduino code for this car circuit

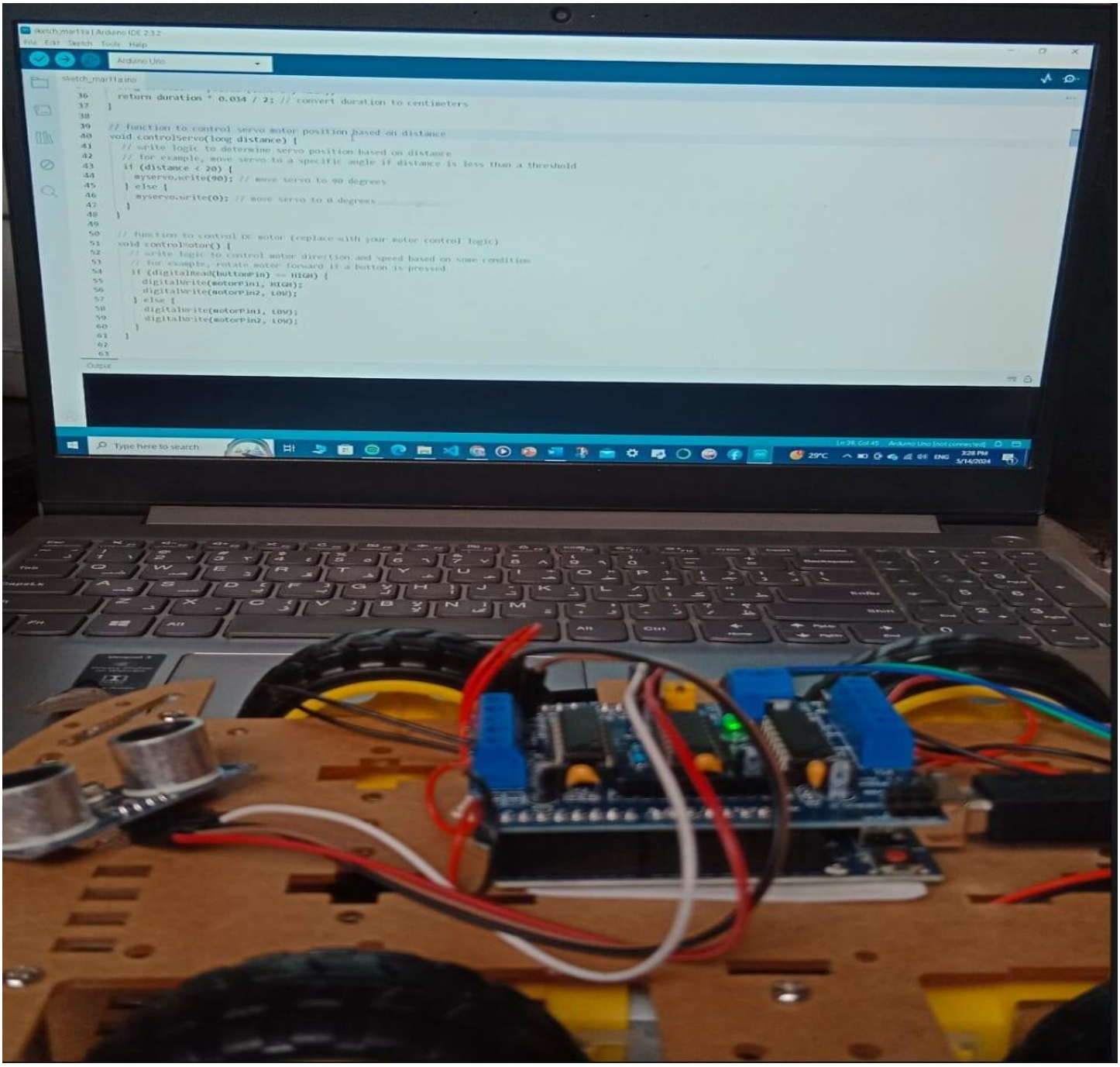


Figure 4. programming

Components for car circuit before connection

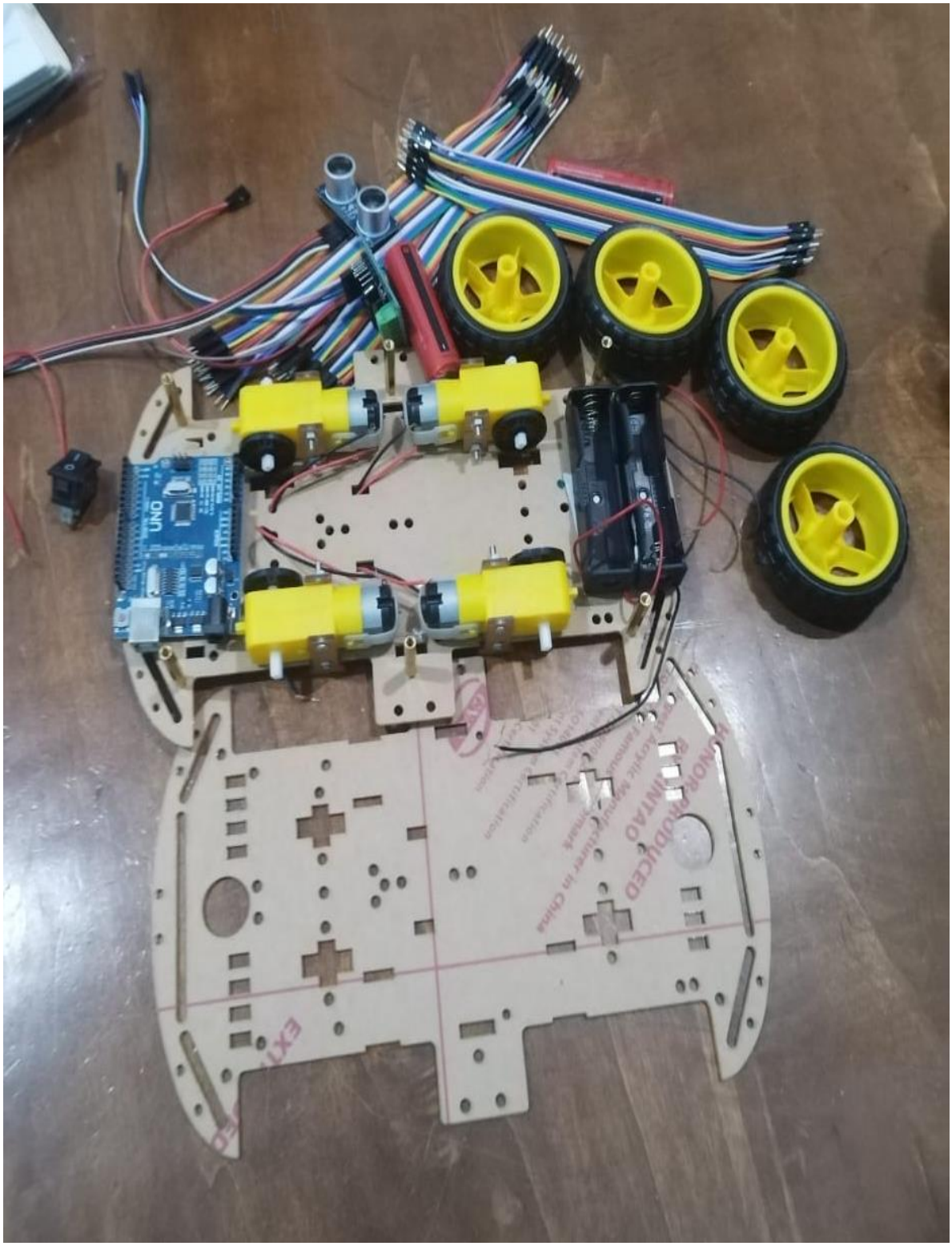


Figure 5. connection

After connection some of components on board

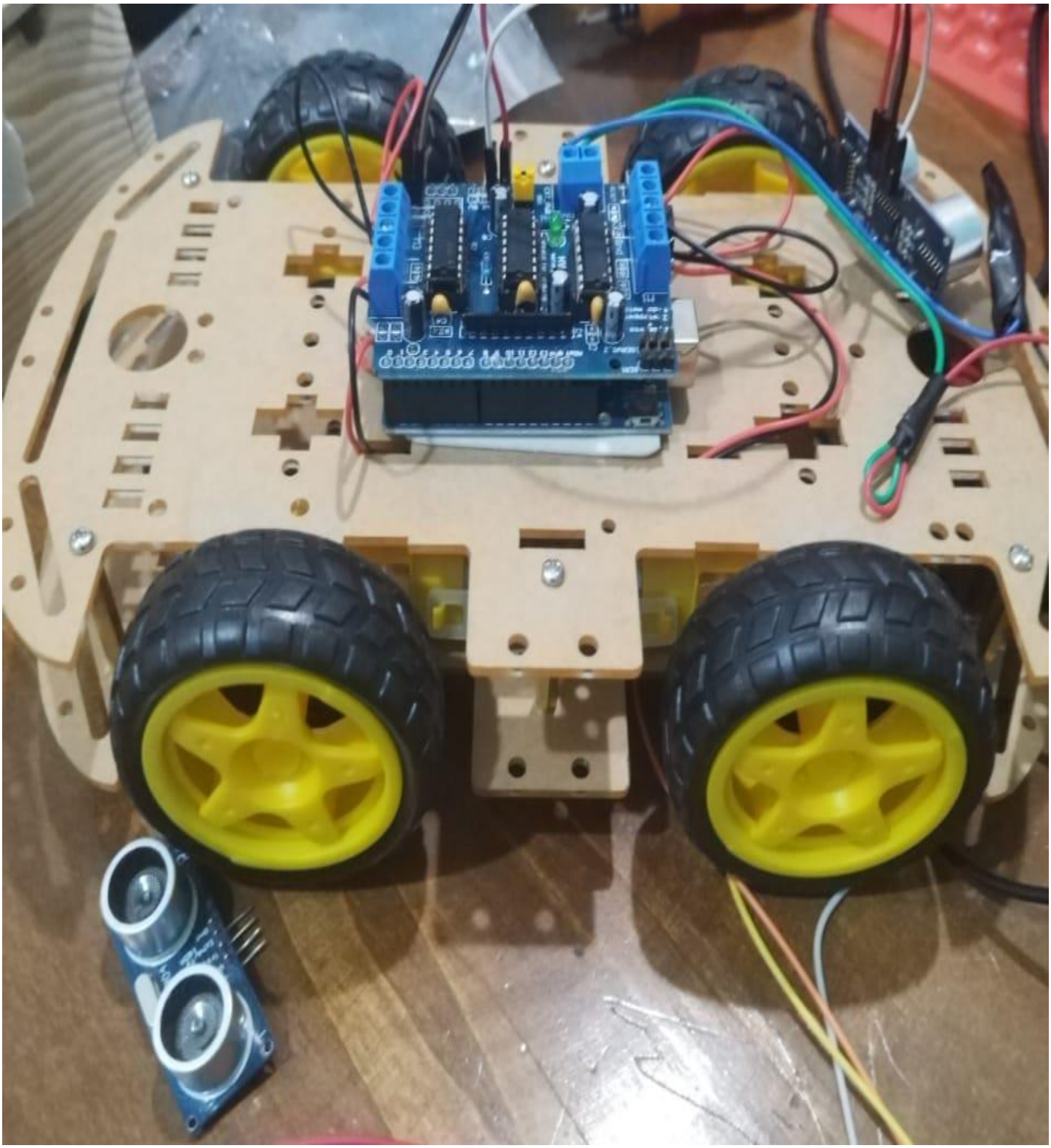


Figure 6. connection

After connection all components for car circuit

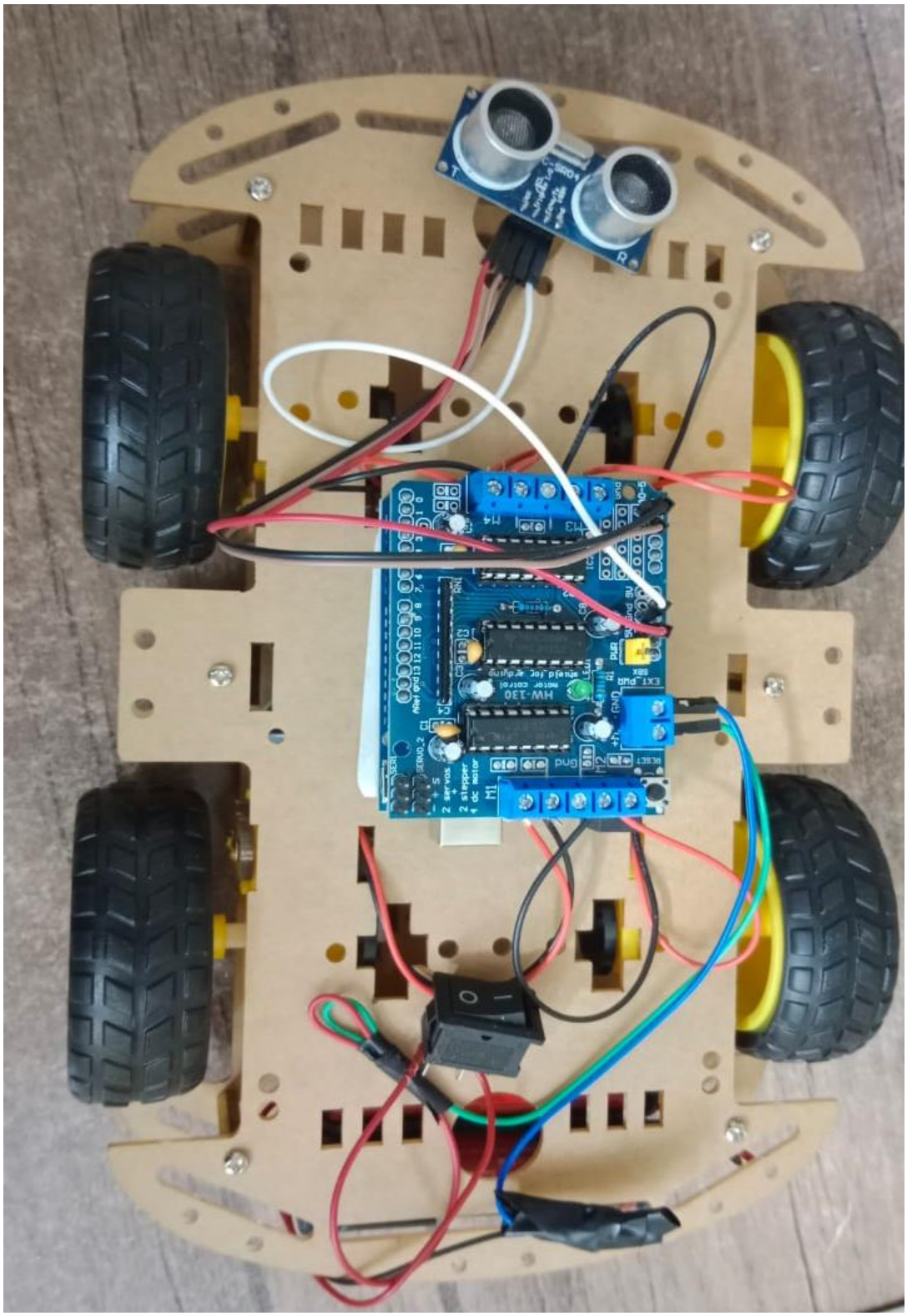


Figure 7. connection

The second part of hardware project

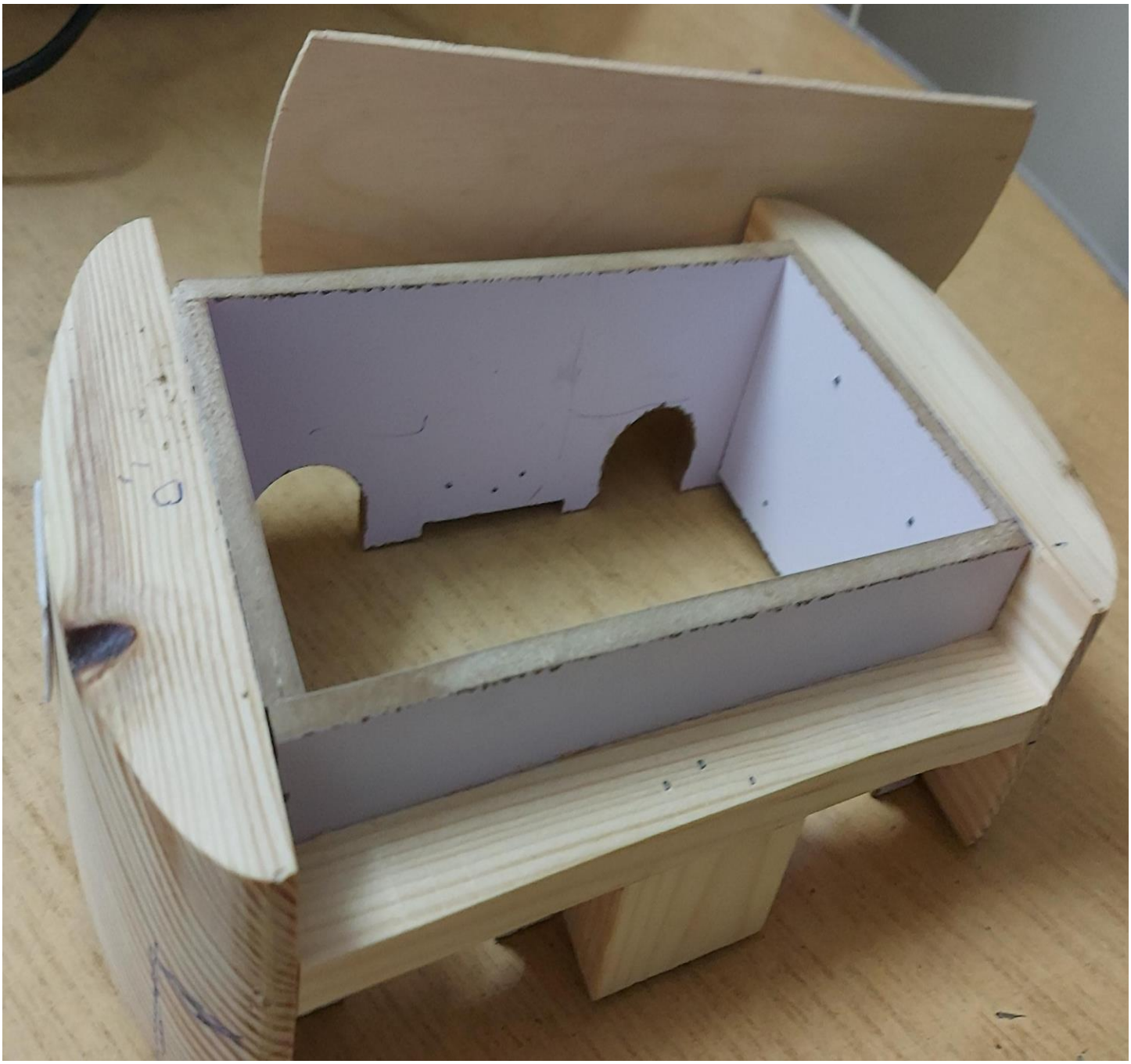


Figure 8. connection

When install the final hardware part of project

Figure 9. final shape of project

The final interface for software part of project

photo

Figure 10. software part of project

The final shape of the project «car circuit» after we made cover

Photo

Figure 11. final shape of project

The final shape of the project «car circuit» after we made cover

photo

Figure 12. front side of project

The final shape of the project «car circuit» after we made cover

Photo

Figure 13. right side of project

The final shape of the project «car circuit» after we made cover

photo

Figure 14. left side of project

6.7 car circuit programming codes:

```
#include <AFMotor.h>
#include <Servo.h>

#define echopin A4 // echo pin
#define trigpin A5 // Trigger pin

Servo myservo;
const int MOTOR_1 = 1;
const int MOTOR_2 = 2;
const int MOTOR_3 = 3;
const int MOTOR_4 = 4;
AF_DCMotor motor1(MOTOR_1, MOTOR12_64KHZ); // create motor object, 64KHz pwm
AF_DCMotor motor2(MOTOR_2, MOTOR12_64KHZ); // create motor object, 64KHz pwm
AF_DCMotor motor3(MOTOR_3, MOTOR12_64KHZ); // create motor object, 64KHz pwm
AF_DCMotor motor4(MOTOR_4, MOTOR12_64KHZ); // create motor object, 64KHz pwm
//=============================================================================
//  Initialization
//=============================================================================

int distance_L, distance_F, distance_R;
long distance;
int set = 20;
void setup() {
  Serial.begin(9600);           // Initialize serial port
  Serial.println("Start");
  myservo.attach(10);
  myservo.write(90);
  pinMode (trigpin, OUTPUT);
  pinMode (echopin, INPUT );
  motor1.setSpeed(180);          // set the motor speed to 0-255
  motor2.setSpeed(180);
  motor3.setSpeed(180);
  motor4.setSpeed(180);
}
```

```
//  Main
void loop() {
 distance_F = data();
 Serial.print("S=");
 Serial.println(distance_F);
  if (distance_F > set){
   Serial.println("Forward");
  motor1.run(FORWARD);         // turn it on going forward
  motor2.run(FORWARD);
  motor3.run(FORWARD);
  motor4.run(FORWARD);
    }
    else{hc_sr4();}
}
long data(){
  digitalWrite(trigpin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigpin, HIGH);
  delayMicroseconds(10);
  distance = pulseIn (echopin, HIGH);
  return distance / 29 / 2;
}
void compareDistance(){
  if (distance_L > distance_R){
  motor1.run(BACKWARD);   // turn it on going left
  motor2.run(BACKWARD);
  motor3.run(FORWARD);
  motor4.run(FORWARD);
    delay(350);
  }
  else if (distance_R > distance_L){
  motor1.run(FORWARD);  // the other right
  motor2.run(FORWARD);
  motor3.run(BACKWARD);
  motor4.run(BACKWARD);
```

```
    delay(350);
  }
  else{
  motor1.run(BACKWARD);  // the other way
  motor2.run(BACKWARD);
  motor3.run(BACKWARD);
  motor4.run(BACKWARD);
   delay(300);
  motor1.run(BACKWARD);   // turn it on going left
  motor2.run(BACKWARD);
  motor3.run(FORWARD);
  motor4.run(FORWARD);
    delay(500);
  }
}
void hc_sr4(){
    Serial.println("Stop");
    motor1.run(RELEASE);         // stopped
    motor2.run(RELEASE);
    motor3.run(RELEASE);
    motor4.run(RELEASE);
    myservo.write(0);
    delay(300);
    distance_R = data();
    delay(100);
    myservo.write(170);
    delay(500);
    distance_L = data();
    delay(100);
    myservo.write(90);
    delay(300);
    compareDistance();
}
```

(Chapter seven)

7.1 Conclusion

Finally, I can conclude this project (car circuit of medical assistant project) is very important because it used to help doctor and patients to identify the diseases and diagnosis and the initial treatment for these patient's symptoms

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References: -

- Function of Arduino uno: https://www.researchgate.net/publication/347408311_A_novel_method_for_Arduino_based_electric_vehicle_emulator#:~:text=The%20Arduino%20controller%20controls%20the,and%20speed%20have%20been%20observed.
- Function of motor drive shield: https://www.robotistan.com/motor-driver-shield-for-arduino#:~:text=Motor%20Driver%20Shield%20is%20a,A%202%20seperate%20step%20motors.
- Function of ultrasonic sensor(HC-SR04): https://www.sciencedirect.com/topics/materials-science/ultrasonic-sensor#:~:text=The%20ultrasonic%20sensor%20is%20a,et%20al.%2C%202020).
- Function of Li-on battery: https://www.sciencedirect.com/science/article/abs/pii/S037877530000402X#:~:text=Lithium%2Dion%20(Li%2Dion)%20batteries%20are%20an%20attractive,number%20of%20charge%E2%80%93discharge%20cycles.
- Function of battery holder: https://uk.rs-online.com/web/content/discovery/ideas-and-advice/battery-holders-guide#:~:text=The%20primary%20function%20of%20a,a%20set%20of%20wire%20leads.
- Function of TT Gear Motor: https://docs.sunfounder.com/projects/ultimate-sensor-kit/en/latest/components_basic/29-component_ttmotor.html#:~:text=A%20TT%20motor%20is%20a,wire%20and%20a%20negative%20wire.
- Function of DC power switch: https://www.researchgate.net/publication/274479326_DC_Power_System_of_Electric_Vehicle#:~:text=Main%20function%20is%20to%20convert,windshield%20wiper%20system%2C%20audio%20system.
- Function of wheels: https://afdc.energy.gov/vehicles/how-do-all-electric-cars-work#:~:text=Electric%20traction%20motor%3A%20Using%20power,the%20drive%20and%20regeneration%20functions.
- Function of male/male wires: https://www.hnhcart.com/products/jumper-wire-male-to-male#:~:text=It%20is%20used%20to%20interconnect,equipment%20or%20components%20without%20soldering.
- Function of female/ male wires: https://blog.sparkfuneducation.com/what-is-jumper-wire#:~:text=The%20difference%20between%20each%20is,used%20to%20plug%20things%20into.
- Connections for car circuit: https://gemini.google.com/app/71ba556ec7841070